Assignment 004: Personal Diary

Program Report

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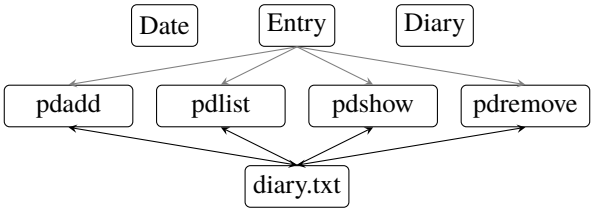
1 Introduction

Personal Diary is a command-line diary made of four programs: pdadd, pdlist, pdshow and pdremove. They share the same classes and keep all entries in one file, diary.txt. Dates are written as YYYYMMDD. The programs were built with g++ 13.3.0 (C++17) on Ubuntu 24.04.

2 Program Design

1. Date stores the year, month and day. It checks whether a date is valid, including leap years, and compares two dates.
2. Entry is one diary entry: a date and its lines of text.
3. Diary keeps all entries in a map sorted by date, so each date has only one entry. It loads and saves diary.txt, where each entry is a date line, its content, and a line with a single ".".

Each program checks its arguments, loads the diary and does its own job. pdadd reads lines until "." or the end of input, and replaces an old entry of the same date. pdlist lists all entries, or only those between two dates. pdshow prints one entry. pdremove returns 0 on success and -1 on failure. Only pdadd and pdremove save the file, because pdlist and pdshow do not change it. Error messages go to stderr, so the output can be redirected.



3 Testing

Each program is compiled with Date.cpp and Diary.cpp, as described in README.md. pd.sh runs a program by its name, for example `bash pd.sh pdshow 20261105`.

`bash pd.sh test` tests adding, listing, replacing, redirection and pipes, removing, and wrong input in a temporary directory. All tests passed.

4 Problems

1. At first parse was a normal member function, so Date::parse could not be called without an object. I made it static.
2. I used `std::map<Date, Entry>` because it keeps the entries sorted by date, so pdlist needs no extra sorting, and it allows only one entry for each date, so pdadd can replace an old entry directly. But map must compare its keys, and Date is my own class, so the compiler did not know which date is earlier. I overloaded `operator<` to compare the year, then the month, then the day. With it, map can sort the entries and find an entry by its date.
3. pdremove returns -1, but the shell shows 255, so the test checks for 255.
4. At first pdshow and pdlist also saved the file. In a pipe like `pdshow A | pdadd B`, both programs run at the same time, and entries were sometimes lost. Now they do not save the file.

5 AI Usage Statement

1. I used AI to learn and review the related knowledge, and to help with a small amount of debugging.
2. I used AI to write the test script.
3. I used AI to add safety checks to the four main programs.

6 Conclusion

The programs can add, list, show and remove diary entries, and work together through redirection, pipes and pd.sh. Build and run instructions are in README.md.